

Matching Starbursts



Note: In bag is 4 red and 2 yellow

The game is simple. Reach your hand into the bag and choose a Starburst. Replace the Starburst into the bag, mix up the bag, then reach in for another. If the color matches the first one, you win! What is the probability of winning this game?

1. Your group should play the game 10 times total.
   a. Number of wins for the group = 6
   b. Based on the results from your group, what is the probability of winning this game? $\frac{6}{10} = 0.60$
2. Let's see what happened in the other groups. Record your number of wins at the front of the room.
   a. Number of wins for the class = 59
   b. Based on the results from your whole class, what is the probability of winning this game?
3. How could we improve this estimate of the probability of winning? $\frac{59}{110} = 0.54$

   Play the game more! Law of Large Numbers

STOP

4. To determine the true probability of winning, we can list out all possible pairs of choices that we could get. Complete the table.

1st candy \ 2nd candy	R	R	R	R	Y	Y
R	RR	RR	RR	RR	RY	RY
R	RR	RR	RR	RR	RY	RY
R	RR	RR	RR	RR	RY	RY
R	RR	RR	RR	RR	RY	RY
Y	YR	YR	YR	YR	YY	YY
Y	YR	YR	YR	YR	YY	YY

sample space

5. Complete the following:
   a. Circle the outcomes with both red. What is the probability of winning with two red?

   $\frac{16}{36} = 0.444$ P(1st red AND 2nd red) $= \frac{4}{6} \times \frac{4}{6} = \frac{16}{36}$
   b. Circle the outcomes with both yellow. What is the probability of winning with two yellow?

   $\frac{4}{36} = 0.111$ P(1st yellow AND 2nd yellow) $= \frac{2}{6} \times \frac{2}{6} = \frac{4}{36}$
   c. Overall, what is the probability of winning?

   $\frac{16}{36} + \frac{4}{36} = \frac{20}{36} = 0.555$ P(win red OR win yellow)

Lesson 3.3 – Rules for Probability

QuickNotes

LT#1 Sample space

List of all possible outcomes

$$P(E) = \frac{\#\text{ of outcomes in } E}{\text{total }\#\text{ outcomes in sample space}}$$

* all outcomes must be equally likely

LT#2 Rules and Notation

Complement rule: $P(A^c) = 1 - P(A)$

probability of an event not happening

$P(A \text{ and } B) = P(A \cap B)$ → both occur

$P(A \text{ or } B) = P(A \cup B)$ → one or the other or both

Check Your Understanding

Pew Research Center reported that family size is shrinking. Suppose we are to randomly select one mother and record the number of children. Here is a probability model.

Number of Children	1	2	3	4+
Probability	0.22	?	0.24	0.13

0.41

(a) Steven argues that the probability of a mother having exactly 2 children is ¼ because there are four categories (1,2,3,4+). Explain why this is incorrect.

Each of the outcomes is not equally likely to occur.

(b) Find the probability of a mother having exactly 2 children. Show your work.

$1 - (0.22 + 0.24 + 0.13) = 1 - 0.59 = 0.41$

For each of the following write the event using proper notation and find the probability:

(c) Find the probability that a randomly selected mother has less than 2 children.

$P(1 \text{ child}) = 0.22$

(d) Find the probability that a randomly selected mother has 1 or 2 children.

$P(1 \text{ child or } 2 \text{ children}) = P(1 \cup 2) = 0.22 + 0.41 = 0.63$

(e) Find the probability that a randomly selected mother does not have 4 or more children.

$P(\text{not } 4+ \text{ children}) = P(4^+ \text{children}^c) = 1 - 0.13 = 0.87$